Binary Jacobian Quotient Representatives and H/G Common-Zero Exclusion

Lean Stage 3 Algebraic Certificates

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Repository: https://github.com/BenFrohman/HODGE
Pull request: https://github.com/BenFrohman/HODGE/pull/38

Status. Version 1.1 of a companion to the Stage 1–2 algebraic notes. It is not peer reviewed and does not claim novelty, a geometric Hodge statement, or anything about the Hodge conjecture. License of the cited Lean sources remains Apache-2.0; this note does not relicense them. Author metadata uses the exact form `Frohman, Benjamin Stanley`.

Abstract

We record thirteen Lean 4 theorems in `Hodge.ExplicitJacobianQuotient` at commit `e56be3d0e848c2098f1d9057dccec324f34bae7f`. Over the actual binary polynomial ring $K[x, y]$, with $f_k = kx^5y + y^6$, the file identifies the Jacobian ideal generated by the two formal partial derivatives, proves the corresponding quotient relations (including $x^4y = 0$ and $y^6 = 0$ when $k \neq 0$), identifies the section Wronskian determinant W with the Stage 2 coefficient formulas, proves the four-monomial identity $[HW] = u_0y^5 + u_1xy^4 + u_2x^2y^3 + u_3x^3y^2$ in the quotient at every modulus root, proves $G^2 - H^3 = f_k$ in the polynomial ring from the six existing coefficient relations, and proves that any common evaluation zero of the actual H and G forces the origin. Combined with Stages 1–2 this yields a GitHub Actions audit of 65 incremental theorem declarations under only `propext`, `Classical.choice`, and `Quot.sound`. This is not a Sylvester-resultant identification, not a quotient basis or dimension theorem, and not a geometric Hodge/residue/cycle-class proof.

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1 Scope and relation to Stages 1–2

This note is a third coherent algebraic paper for a distinct substantive result: the actual binary Jacobian-quotient representatives and the projective common-zero exclusion for the section forms H, G. Supporting lemmas that are solely scaffolding for that result are bundled here; Stages 1 and 2 remain separate papers for the determinant/derivative identities and the Bézout nonvanishing certificates [1].

Source. Module `Hodge/ExplicitJacobianQuotient.lean` at `e56be3d0e848c2098f1d9057dccec324f34bae7f` (PR https://github.com/BenFrohman/HODGE/pull/38). Byte SHA-256 of the Lean source: `0ff4a38ce58504ad4e4b53f327665e690895c795db34ebcbbd99159ce97e6ae5` (matches `stage3-manifest.json`).

What this note proves. Exactly the thirteen theorems listed in Section 4, with hypotheses taken from the Lean declarations (no hidden assumptions).

What this note does not prove.

- Identification of mathlib's Sylvester resultant, or a Lean check of the tenth Python numeric resultant/Bézout certificate.
- Canonical normal forms, a basis of the whole quotient, or quotient dimension. (That the four degree-five classes are linearly independent, and hence that $[HW] \neq 0$, is proved in the companion milestone note `milestones/degree-five-independence/main.pdf`, module `Hodge/ExplicitJacobianIndependence.lean`; it is not part of this note.)
- Reducedness/saturation of geometric ideals, schemes, Betti/Hodge comparison, residue-cycle theorems, Galois/Fourier descent, or any general/integral Hodge conjecture statement.

Terminology: the binary determinant W below is the Wronskian of the section forms H, G (formal partials). It is *not* Jacobi's identity for Lie brackets, and it is not a claim that W is a cohomology class.

2 Standing definitions (exact Lean objects)

Throughout, K is a commutative ring, a field, or a characteristic-zero field with a $\mathbb{Q}$-algebra structure, as required by each theorem.

Definition 1 (Binary ring and f_k)**.** `Binary K` = `MvPolynomial (Fin 2) K`, with $x = X(0)$, $y = X(1)$, and

$$f_k \;=\; k\,x^5y + y^6 \qquad (\texttt{binarySextic}\ k). \tag{1}$$

This is the scaled section polynomial; the unscaled $x^5y + y^6$ is not substituted.

Definition 2 (Jacobian ideal and quotient)**.**

$$J(k) \;=\; (\partial_x f_k,\ \partial_y f_k) \qquad (\texttt{jacobianIdeal}\ k), \tag{2}$$
$$\mathrm{Jac}(k) \;=\; K[x,y]/J(k) \qquad (\texttt{Jacobian}\ k). \tag{3}$$

Write $\overline{x}, \overline{y}$ for the images of x, y (`qx`,`qy`), and $c \mapsto \overline{c}$ for the constant map (`constants`).

Definition 3 (Section forms)**.** With Stage 2/ExplicitCurve coefficients $b = \mathrm{sb}(k)$, …, $g = \mathrm{sg}(k)$,

$$H \;=\; x^2 + b\,xy + c\,y^2, \tag{4}$$
$$G \;=\; x^3 + d\,x^2y + e\,xy^2 + g\,y^3, \tag{5}$$
$$W \;=\; \partial_x H \cdot \partial_y G - \partial_y H \cdot \partial_x G. \tag{6}$$

These are `sectionH`, `sectionG`, `sectionW`. The Stage 2 scalars w_i, u_i are exactly the certificates `w0`,…,`u3` from `ExplicitResidueCertificates`.

Definition 4 (Modulus-root hypothesis)**.**

$$k^{24} + 66081312000\,k^{18} - 4811512860000\,k^{12} + 1167566400000\,k^6 + 583200000 = 0. \tag{7}$$

Irreducibility of this polynomial over $\mathbb{Q}$ is neither assumed nor proved.

3 Main results

Theorem 5 (Partial derivatives of f_k)**.** *For a commutative ring K and $k \in K$,*

$$\partial_x f_k = 5k\,x^4y, \qquad \partial_y f_k = k\,x^5 + 6y^5. \tag{8}$$

Lean. `partial_x`, `partial_y`: unfold and ring/simp. □

Theorem 6 (Generators vanish in the quotient)**.** *In* $\mathrm{Jac}(k)$*,*

$$\overline{5k}\,\overline{x}^4\overline{y} = 0, \qquad \overline{k}\,\overline{x}^5 + 6\,\overline{y}^5 = 0. \tag{9}$$

In particular the second relation is $\overline{k}\,\overline{x}^5 = -6\overline{y}^5$, not $\overline{x}^5 = -6\overline{y}^5$.

Lean. `partial_x_class`, `partial_y_class`: membership of the generators in $J(k)$ and the quotient zero criterion. □

Lemma 7 (Constant inverse)**.** *If K is a field and $a \in K \setminus \{0\}$, then $\overline{a^{-1}}\,\overline{a} = 1$ in* $\mathrm{Jac}(k)$*.*

Lean. `constant_inverse`. □

Theorem 8 (Monomial vanishings for $k \neq 0$)**.** *Assume K is a field of characteristic zero and $k \neq 0$. Then in* $\mathrm{Jac}(k)$*,*

$$\overline{x}^4\overline{y} = 0 \qquad \textit{and} \qquad \overline{y}^6 = 0. \tag{10}$$

Lean. `x4y_class`: invert $\overline{5k}$ using Lemma 7 and Theorem 6. `y6_class`: combine $\overline{y} \cdot (\partial_y$-class) with $\overline{k}\,\overline{x} \cdot (\overline{x}^4\overline{y} = 0)$ and invert 6. □

Theorem 9 (Wronskian equals Stage 2 W-coefficients)**.** *Assume K is a characteristic-zero field with a $\mathbb{Q}$-algebra structure. Then*

$$W = C(w_3)\,x^3 + C(w_2)\,x^2y + C(w_1)\,xy^2 + C(w_0)\,y^3 \tag{11}$$

in $K[x,y]$, where each $w_i \in K$ is the Stage 2 scalar coefficient and $C\colon K \to K[x,y]$ embeds constants. The quotient image in $\mathrm{Jac}(k)$ *retains bars only on the quotient representatives:*

$$\overline{W} = \overline{C(w_3)}\,\overline{x}^3 + \overline{C(w_2)}\,\overline{x}^2\overline{y} + \overline{C(w_1)}\,\overline{x}\,\overline{y}^2 + \overline{C(w_0)}\,\overline{y}^3. \tag{12}$$

Lean. `wronskian_polynomial` applies Stage 1 `binary_wronskian` at the actual section coefficients; `wronskian_class` pushes forward to the quotient. □

Theorem 10 (H class)**.** *Under the same field/$\mathbb{Q}$-algebra hypotheses,*

$$\overline{H} = \overline{x}^2 + \overline{b}\,\overline{x}\,\overline{y} + \overline{c}\,\overline{y}^2. \tag{13}$$

Lean. `sectionH_class`. □

Theorem 11 (Four-monomial $[HW]$ identity)**.** *Assume K is a characteristic-zero field with a $\mathbb{Q}$-algebra structure and that k satisfies* (7)*. Then in* $\mathrm{Jac}(k)$*,*

$$\overline{HW} = \overline{u_0}\,\overline{y}^5 + \overline{u_1}\,\overline{x}\,\overline{y}^4 + \overline{u_2}\,\overline{x}^2\overline{y}^3 + \overline{u_3}\,\overline{x}^3\overline{y}^2, \tag{14}$$

with the Stage 2 definitions $u_0 = cw_0 - 6$, $u_1 = bw_0 + cw_1$, $u_2 = w_0 + bw_1 + cw_2$, $u_3 = w_1 + bw_2 + ck$. The constant -6 arises from the kx^5 term together with the actual ∂_y relation; it is not obtained by setting $k = 1$.

Proof. Write $H = x^2 + bxy + cy^2$ and $W = w_3x^3 + w_2x^2y + w_1xy^2 + w_0y^3$. Multiplying out,

$$\begin{aligned} HW = {} & w_3x^5 + (w_2 + bw_3)x^4y + (w_1 + bw_2 + cw_3)x^3y^2 \\ & + (w_0 + bw_1 + cw_2)x^2y^3 + (bw_0 + cw_1)xy^4 + cw_0y^5. \end{aligned}$$

In $\mathrm{Jac}(k)$, $\overline{x}^4\overline{y} = 0$ (Theorem 8, which needs $k \neq 0$), and $\overline{k}\,\overline{x}^5 = -6\overline{y}^5$ (Theorem 6). Since $w_3 = k$ (`ExplicitCurve.rel_h5`), the x^5 term becomes $-6\overline{y}^5$, the x^4y term drops, and collecting gives $u_0 = cw_0 - 6$, $u_1 = bw_0 + cw_1$, $u_2 = w_0 + bw_1 + cw_2$, $u_3 = w_1 + bw_2 + ck$. Lean: `section_HW_class`: use $w_3 = k$ via `ExplicitCurve.rel_h5`, multiply the H and W classes, and reduce with Theorems 8 and 6 ($k \neq 0$ from `section_root_ne_zero`). □

Theorem 12 (Section identity in the polynomial ring)**.** *Under the same hypotheses as Theorem 11,*

$$G^2 - H^3 = f_k \tag{15}$$

as an equality in $K[x,y]$, derived from the six existing checked coefficient relations (`rel_h5` through `rel_h0`) and (7)*, not postulated.*

Lean. `section_identity_polynomial`: push the six scalar relations into constants and apply `ExplicitCurve.section_identity`. □

Theorem 13 (Common-zero exclusion)**.** *Under the same hypotheses as Theorem 11, if $a, b \in K$ satisfy*

$$H(a,b) = 0 \qquad \text{and} \qquad G(a,b) = 0, \tag{16}$$

then $a = 0$ *and* $b = 0$.

Proof. `section_common_zero`. Differentiate Theorem 12: $\partial f_k = 2G\,\partial G - 3H^2\,\partial H$ for $\partial = \partial_x, \partial_y$. At a common zero of H and G both terms vanish, so both partials of f_k vanish:

$$5ka^4b = 0, \qquad ka^5 + 6b^5 = 0. \tag{17}$$

From (7) one has $k \neq 0$ (its constant term 583200000 is nonzero), and $5 \neq 0$ in characteristic zero, so $a = 0$ or $b = 0$. If $a = 0$ the second equation is $6b^5 = 0$, so $b = 0$; if $b = 0$ it is $ka^5 = 0$, so $a = 0$. No resultant API, coprimality premise, or external geometric theorem is assumed. This discharges the projective common-zero input supplied classically by the tenth Python certificate, without identifying Sylvester resultants. □

4 Theorem–declaration mapping

#	Lean declaration	Content
1	`partial_x`	Thm. 5: $\partial_x f_k$
2	`partial_y`	Thm. 5: $\partial_y f_k$
3	`partial_x_class`	Thm. 6: $\overline{5k}\,\overline{x}^4\overline{y} = 0$
4	`partial_y_class`	Thm. 6: $\overline{k}\,\overline{x}^5 + 6\overline{y}^5 = 0$
5	`constant_inverse`	Lem. 7
6	`x4y_class`	Thm. 8: $\overline{x}^4\overline{y} = 0$
7	`y6_class`	Thm. 8: $\overline{y}^6 = 0$
8	`wronskian_polynomial`	Thm. 9: W coefficients
9	`wronskian_class`	Thm. 9: $\overline{W}$
10	`sectionH_class`	Thm. 10
11	`section_HW_class`	Thm. 11: four-monomial $[HW]$
12	`section_identity_polynomial`	Thm. 12: $G^2 - H^3 = f_k$
13	`section_common_zero`	Thm. 13: only origin

Namespace: `Hodge.ExplicitJacobianQuotient`. Supporting lemmas for this result are kept in this paper; Stages 1–2 are not re-split here.

5 Verification evidence (GitHub Actions)

The Lean file `Hodge/ExplicitJacobianQuotient.lean` is byte-identical at `e56be3d`, at `main` `b13f57f`, and at `52325b02f966139c99015443ca5fba8a725ef3bb` (pull request #49), where it was checked again. In that run the workflow `certify.yml` (https://github.com/BenFrohman/HODGE/actions/runs/42922528702, passed) built the residue modules and printed, for each of the thirteen theorems of this note,

```
'Hodge.ExplicitJacobianQuotient.partial_x'
'Hodge.ExplicitJacobianQuotient.partial_y'
'Hodge.ExplicitJacobianQuotient.partial_x_class'
'Hodge.ExplicitJacobianQuotient.partial_y_class'
```

```
'Hodge.ExplicitJacobianQuotient.constant_inverse'
'Hodge.ExplicitJacobianQuotient.x4y_class'
'Hodge.ExplicitJacobianQuotient.y6_class'
'Hodge.ExplicitJacobianQuotient.wronskian_polynomial'
'Hodge.ExplicitJacobianQuotient.wronskian_class'
'Hodge.ExplicitJacobianQuotient.sectionH_class'
'Hodge.ExplicitJacobianQuotient.section_HW_class'
'Hodge.ExplicitJacobianQuotient.section_identity_polynomial'
'Hodge.ExplicitJacobianQuotient.section_common_zero'
  each: depends on axioms: [propext, Classical.choice, Quot.sound]
PASS: 78 incremental theorem declarations; standard axioms only
```

($78 = 7 + 45 + 13 + 13$: Stages 1–3 and the degree-five milestone). The same run's root audit printed `audited 699 declarations, 699 with standard axioms only`, and `scripts/check_no_sorry.py` passed. The library build and the CI sorry scan, which now lists this file, passed in https://github.com/BenFrohman/HODGE/actions/runs/42922528652. A kernel check shows the Lean statements follow from the Lean definitions; it does not establish any geometric statement.

6 Toolchain pins and reproducibility

Pin	Value
Lean	`leanprover/lean4:v4.22.0`
mathlib input	`v4.22.0`
mathlib checkout	`79e94a093aff4a60fb1b1f92d9681e407124c2ca`
Source commit	`e56be3d0e848c2098f1d9057dccec324f34bae7f` (PR #38)
Re-checked at	`52325b02f966139c99015443ca5fba8a725ef3bb` (PR #49)
License	Apache-2.0 (preserved; no relicense)

One command from the repository root:

```
lake exe cache get && lake build && python3 scripts/check_residue_axioms.py
```

7 Limitations

Theorem 11 exhibits a representative of $\overline{HW}$ with coefficients u_i. Nonvanishing of those scalars (Stage 2) does *not* by itself prove that the class is nonzero; that needs the independence of the four degree-five classes, which is proved separately in the degree-five milestone note. A basis or dimension theorem for the whole quotient is still not formalized. Theorem 13 excludes common nonzero zeros of H, G but does not formalize Proj, schemes, or residue maps. Villaflor Loyola's residue-cycle theorem [2] and Griffiths comparison remain external classical input.

Author, license, and contact

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References

[1] Benjamin Stanley Frohman. HODGE: Lean 4 formalization and hodge ledger. https://github.com/BenFrohman/HODGE, 2026. Source commit e207394a9ed4255ad804860b8f69fde7e940dd54; PR https://github.com/BenFrohman/HODGE/pull/36. Historical archive DOI 10.5281/zenodo.23629480 identifies prior repository provenance and is not a DOI for these incremental algebra notes.

[2] Roberto Villaflor Loyola. Periods of complete intersection algebraic cycles. *Manuscripta Mathematica*, 167:765–792, 2022. Cited as classical residue-cycle input; not formalized in the incremental Lean modules of this note.